A Multi-Agent Framework for Safe Hospital-to-Home Transitions

(Phase 1: Track B)

Team Members: Anshika Shukla, Nicole Sanna, Nicole Zhou, Yuhang Zhou

I. Problem Statement

Current post-discharge care relies heavily on patients to manage complex instructions, monitor symptoms, and coordinate follow-up care on their own. These are tasks especially difficult for elderly individuals, patients with chronic conditions, and those with low health literacy or limited support. As a result, early warning signs are often missed, medication errors occur, and care gaps lead to preventable complications. Approximately **one in five** patients experience adverse events during this transition, including unplanned readmissions within a month of discharge, medication errors, and even death. [1] Existing systems are largely reactive and fragmented, lacking continuous monitoring and timely intervention. There is a clear need for a proactive, coordinated support system that can guide patients, track their condition over time, and enable early, targeted interventions to improve outcomes and reduce healthcare burden.

II. Project Summary

Hospital-to-home transitions are a critical failure point in healthcare, where patients often mismanage medications, miss follow-ups, or fail to recognize early warning symptoms, leading to preventable readmissions. This project proposes a multi-agent system, called the ***Post-Discharge Care Companion***. This system actively supports patients during recovery by translating discharge instructions into actionable plans, continuously monitoring symptoms, and coordinating with healthcare and insurance systems. By combining planning, reasoning, and real-time interaction, the system serves as a personalized, proactive care companion that reduces patients' cognitive burden and improves post-discharge outcomes.

III. Why is this an Agentic Problem?

This problem is agentic because it requires continuous, stateful decision-making over time, not one-time responses. The system must track patient history, adapt to new information, and take proactive actions toward improving outcomes. It also coordinates multiple tasks and escalates to humans when needed, making it inherently agentic.

It has a persistent memory and continuously tracks states. The system keeps track of the patient's daily progress, such as medications taken, missed doses, and recurring symptoms (e.g., dizziness

[1] Hardicre NK, Birks Y, Murray J, Sheard L, Hughes L, Heyhoe J, et al. Partners at care transitions (PACT) -e xploring older peoples' experiences of transitioning from hospital to home in the UK: protocol for an observation and interview study of older people and their families to understand patient experience and involvement in care at transitions. BMJ Open. (2017) 7:e018054. 10.1136/bmjopen-2017-018054

over several days). By remembering past information, it can recognize patterns and identify when something is getting worse, rather than treating each update as an isolated event.

Instead of asking the patient to search on their own, the system can automatically connect to external services, such as provider directories or insurance information, to find in-network doctors, estimate costs, and suggest the next steps, reducing effort and confusion. It is an intrinsically coordinated system, with different parts working together like a team. For instance, one agent may detect a potential issue, another updates the patient's care plan or reminders, and another helps arrange follow-up care. This coordination allows the system to handle complex situations more effectively than a single tool.

The system is Autonomous and doesn't wait for the user to take action. It's goal driven behavior actively checks in, sends reminders, and adjusts recommendations over time to help the patient stay on track and reduce risks, especially during the critical recovery period. Lastly, the system accounts for a human-in-the-loop for its escalation mechanisms. When the system detects something potentially serious, it alerts a real person, such as a nurse or caregiver, to step in and make a more informed decision, ensuring safety and trust.

IV. Target User, Stakeholder, and Audience

The system primarily targets patients who are at **high risk during post-discharge recovery**, especially those with complex or ongoing care needs. These include:

Primary Users: Patients

- **Elderly patients with chronic illnesses:** Patients with heart failure, diabetes, or COPD, who often manage multiple medications and require continuous monitoring. These patients are particularly vulnerable to medication errors and delayed responses to worsening symptoms.
- **Post-surgical patients**: Patients who are recovering from cardiac procedures or joint replacements must follow strict recovery protocols, including medication schedules, physical therapy, and follow-up visits.
- **Patients with low health literacy or language barriers**: Patients who may struggle to interpret discharge documents and instructions, increasing the likelihood of non-adherence.
- **Patients living alone or without strong social support**: These patients lack immediate assistance in recognizing or responding to health risks after discharge.

Secondary Users: Caregivers

Caregivers play a critical supporting role in ensuring patient safety and adherence. This group includes:

- **Family members (e.g., spouses, adult children):** They can help monitor the patient's condition and assist with daily care decisions
- **Home health aides and visiting nurses:** They provide in-home support and may rely on the system to track patient status and respond to alerts.

Clinical Stakeholders: Healthcare Providers

Healthcare professionals are involved as **human-in-the-loop decision-makers**, particularly when the system detects elevated risk. This group includes:

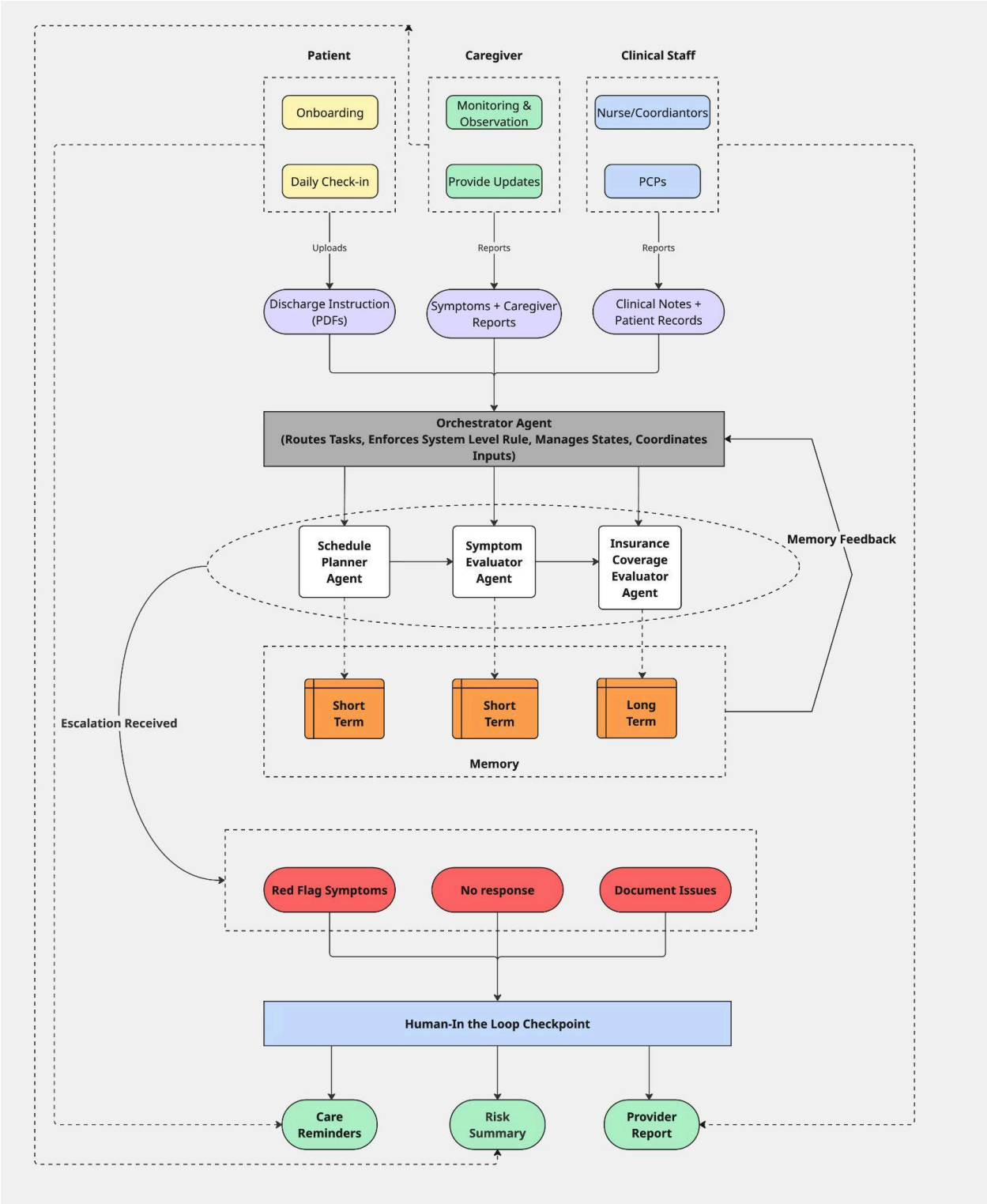
- **Nurses and care coordinators**, who respond to alerts generated by the system and provide timely interventions
- **Primary care physicians and specialists**, who use summarized patient data to inform follow-up care decisions

Institutional Stakeholders: System Adopters

At the organizational level, the system is most likely to be adopted and deployed by:

- **Hospitals and health systems**, particularly discharge planning teams aiming to reduce readmission rates
- **Health insurers and payers**, who are financially incentivized to minimize avoidable hospitalizations

V. Initial Architecture Concept



VI. Initial Success Criteria

The success of the system will be evaluated based on its ability to improve post-discharge care and reduce risk through timely, coordinated support. Key criteria include:

- **Early risk detection:** The system identifies potential complications (e.g., concerning symptom patterns) earlier than standard care processes.
- **Improved adherence:** Patients follow medication schedules and care plans more consistently.
- **Timely escalation:** High-risk cases are appropriately flagged and escalated to human providers.
- **Reduction in preventable readmission risk:** Demonstrated through simulation or case-based evaluation rather than real clinical outcomes.

VII. Scope Boundaries

The project focuses on building a functional prototype that demonstrates the core capabilities of a multi-agent post-discharge support system. Within scope, the system will parse discharge instructions into structured and actionable care plans, enable daily symptom check-ins, and perform basic risk assessment using rule-based or simple model-driven logic. It will also simulate alerting and escalation mechanisms. The primary goal is to showcase how multiple agents can coordinate effectively within a controlled, simulated environment.

Out of scope are features that require real-world clinical deployment or regulatory compliance. The system will not perform full clinical-grade diagnosis or medical decision-making, nor will it implement standards such as HIPAA. Additionally, complex insurance workflows, such as billing or claims processing, are excluded. These limitations allow the project to remain focused on demonstrating the agentic architecture and core functionality without the overhead of real-world operational constraints.

In Scope:

- Patient Information Database creation/simulation
- Parsing discharge instructions into structured care plans
- Daily symptom check-ins and basic risk assessment
  Rule-based or simple model-driven alerting and escalation
- Integration with limited external data sources (e.g., mock provider/insurance data)
  Prototype or simulation of multi-agent workflows
- Human intervention mechanism

Out of Scope:

- Full clinical-grade diagnosis or decision-making
- Regulatory compliance (e.g., HIPAA implementation, PHI Regulations)
- Complex insurance billing, claims processing, or pre-auth scenarios. We will be building a happy path scenario simulator.

VIII. One Page "Multi-agent System Canvas"

HOSPITAL-TO-HOME MULTI-AGENT SYSTEM CANVAS

1 SYSTEM INTENT

User Case / Purpose

Support patients during the hospital-to-home transition by translating discharge instructions into actionable recovery plans, monitoring symptoms over time, and helping coordinate follow-up care. The purpose is to reduce patient confusion, improve adherence, and surface early warning signs before they become preventable complications.

Task Slice / Goal

We aim to build a prototype post-discharge support experience that helps patients understand discharge instructions, follow a daily recovery plan, complete symptom check-ins, and receive escalation guidance when potential red flags appear.

Users / Stakeholders

REQUESTER	High-risk patients (elderly, post-surgical) & Caregivers
VIEWER	Triage Nurses, Care Coordinators
DOWNSTREAM	Primary Care Physicians, Hospitals, Insurers (ACOs)

2 BOUNDED BEHAVIOR

Input / Output

IN	Discharge instruction, Medication list, Follow-up care recommendation, Daily symptom reports from patient, Patient recovery history, Insurance information
OUT	Structured daily care plan, Medication and recovery reminders, Symptom risk assessment, Escalation recommendation, Follow-up care guidance, In-network provider suggestions, Summarized patient status for human review

Tools / Permissions

ALLOW	Document parsing tool for discharge instruction, Calendar API for mock scheduling and reminding, Provider directory lookup tool
DENY	Generating clinical-grade medical diagnoses, Modify official EHR (Electronic Health Records), Process real medical claims

Autonomy / Human Checkpoints

AUTONOMOUS	Extract instructions, Send routine reminders, Access insurance databases, Run symptom check-ins, Suggest next steps
GATED	Detection of "Red-Flag" symptoms —— Escalates to Human Nurse for clinical intervention

Operating Rules / Policies

HARD	Never provide definitive medical advice; always frame as a recommendation to seek human medical attention
SOFT	Maintain an empathetic, accessible, and jargon-free tone suitable for patients with low health literacy
ORG	Operate as a simulated prototype exempt from clinical-grade regulatory compliance (HIPAA, PHI) and complex billing integrations

3 CONTEXT & EVALUATION

Memory / Context

SESSION	Current daily check-in conversation and symptom inputs
PERSIST	Parsed care plan, longitudinal symptom history (e.g., tracking dizziness over 3 consecutive days), adherence logs
LIMIT/ PRIVACY	Data context is strictly bounded to the 30-day simulated post-discharge window, No real HIPAA data utilized

Failure Conditions / Escalations

RED FLAG	Patient reports severe symptoms —— Immediately route to emergency protocols / Human-in-the-loop
NO RESPONSE	Patient misses >3 consecutive check-ins —— Alert primary caregiver
SOURCE GAP	Unparsable discharge document or key follow-up dates missing —— Prompt user or human discharge planner for clarification
TOOL DOWN	Retry once, then escalate

Success Criteria

EARLY DETECTION	Successfully identify concerning symptom trajectories at least 24 hours earlier than standard manual follow-up workflows
QUALITY	Agent A must achieve near 100% extraction accuracy for critical entities (e.g., medication names, dosages, frequencies) from discharge PDFs Agent C must yield 0 hallucinations when retrieving and recommending in-network providers
ACCURACY	The system must minimize false negatives while actively controlling false positives to prevent alert fatigue among human reviewers
ENGAGEMENT	Achieve and maintain a high completion rate for patient interactive daily check-ins over the targeted post-discharge period.
SAFETY	No unauthorized autonomous clinical decision-making. 100% of defined high-risk or low-confidence cases must be successfully escalated to a human reviewer

IX. One page "Agent Canvas" for each agent or component

Agent System Specification Canvas

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Agent A - Schedule Planner: Converts discharge instructions into a dynamic, personalized care calendar

1 SYSTEM INTENT

Use Case / Purpose (1)

Parse and structure discharge documents into a dynamic, day-by-day care calendar covering medication schedules, physical therapy sessions, and follow-up appointments, reducing cognitive burden on patients during post-discharge recovery.

Task Slice / Goal (2)

Given a set of discharge instructions (uploaded or pasted text), extract all medications with dosages and timing, physical therapy exercises with frequency, and scheduled follow-up visits, then produce a structured daily/weekly care plan with reminders.

Users / Stakeholders (3)

PRIMARY	Post-discharge patient (receives structured care calendar and daily medication reminders)
SECONDARY	Caregivers and family members managing recovery
DOWNSTREAM	Agent B (receives med list for side-effect checks); Agent C (receives follow-ups for coverage lookup)

2 BOUNDED BEHAVIOR

Inputs / Outputs (4)

IN	Discharge instruction document (uploaded text or pasted content)
IN	Patient preferences (e.g., preferred reminder times, language)
OUT	Structured daily/weekly care calendar with medication times, PT exercises, and follow-up dates
OUT	Timed reminders (medication doses, PT sessions, upcoming appointments) sent to patient
OUT	Medication list shared to Agent B; follow-up appointment list shared to Agent C

Tools / Permissions (5)

ALLOW	Document parser (extract text from discharge papers)
ALLOW	Medication reference lookup (dosage validation, interaction flags)
ALLOW	Calendar/schedule generator; reminder system
DENY	Modifying original discharge instructions; prescribing or changing medications
DENY	Rescheduling or canceling follow-up appointments without explicit patient/provider consent
DENY	Sharing patient data outside the care team or in ways that violate HIPAA

BOUNDED AGENTIC SYSTEM™

Autonomy / Human Checkpoints (6)

AUTONOMOUS	Parse document, extract medications and appointments, generate schedule
AUTONOMOUS	Send structured med list to Agent B and follow-up list to Agent C
GATED	Patient or caregiver confirms the generated schedule before activation
GATED	Ambiguous or conflicting instructions → flag for human review

Operating Rules / Policies (7)

HARD	Never add, remove, or alter prescribed medications or dosages
HARD	Never fabricate instructions not present in the source document
HARD	Never reschedule or cancel follow-up appointments autonomously
HARD	Comply with HIPAA and applicable healthcare data protection regulations at all times
SOFT	Use plain language; simplify medical jargon for readability
SOFT	Flag potential drug interactions but do not make clinical judgments

3 CONTEXT & EVALUATION

Memory / Context (8)

SESSION	Raw discharge text, extracted medication/appointment data, draft schedule
PERSIST	Finalized care plan (shared with Agent B and Agent C); patient preferences
CONTEXT	Medication reference data for dosage validation and interaction checks

Failure Conditions / Escalations (9)

FAIL	Discharge document is illegible, incomplete, or cannot be parsed
FAIL	Generated schedule contradicts or misrepresents the source discharge instructions
ESCALATE	Potential drug interaction detected between two prescribed medications → flag for provider/pharmacist
ESCALATE	Unrecognized medication name or dosage outside normal range → flag for nurse/pharmacist
ESCALATE	Patient or caregiver disputes a schedule item → route to care coordinator

Success Criteria (10)

COMPLETENESS	All medications, PT exercises, and follow-up appointments from discharge papers are captured in the schedule
ACCURACY	Zero fabricated instructions; all schedule items traceable to source document
CLARITY	Schedule is understandable by a patient with low health literacy
HANDOFF	Medication list successfully shared to Agent B; follow-up list shared to Agent C
SAFETY	All ambiguities and potential conflicts flagged rather than silently resolved

Post-Discharge Care Guardian - Agent A of 3

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Agent System Specification Canvas

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Agent B - Symptom Evaluator: Daily patient check-ins with rule-based risk assessment and nurse escalation

1 SYSTEM INTENT

Use Case / Purpose (1)

Conduct daily symptom check-ins with post-discharge patients using structured prompts and open-ended follow-ups, assess reported symptoms against the patient's medication profile and known risk factors, and escalate to a human nurse when red-flag patterns are detected, with the goal of catching complications earlier than standard care.

Task Slice / Goal (2)

Each day: prompt the patient with structured symptom questions, allow free-text elaboration, cross-reference responses against the medication list from Agent A, apply rule-based severity logic, and either reassure the patient, recommend self-care, or alert a nurse. If a follow-up visit is recommended, notify Agent C to check coverage.

Users / Stakeholders (3)

PRIMARY	Post-discharge patient (completes daily symptom check-ins via structured prompts and free-text)
DOWNSTREAM	Nurse or care coordinator (receives alerts when risk is elevated)
UPSTREAM	Agent A (provides medication list and known side effects)
DOWNSTREAM	Agent C (triggered when a follow-up visit is recommended)

2 BOUNDED BEHAVIOR

BOUNDED AGENTIC SYSTEM

Inputs / Outputs (4)

IN	Structured symptom responses (pain level, dizziness, swelling, etc.)
IN	Open-ended patient text (free description of how they feel)
IN	Medication list and side-effect profile from Agent A
OUT	Risk assessment (green / yellow / red) with reasoning
OUT	Nurse alert (if red); self-care guidance (if green/yellow); follow-up referral to Agent C (if visit recommended)

Tools / Permissions (5)

ALLOW	Symptom questionnaire engine (structured prompts + free-text intake)
ALLOW	Medication side-effect reference (cross-check symptoms against med profile)
ALLOW	Rule-based severity classifier; alert/notification system
DENY	Diagnosing conditions; prescribing treatment; changing medications
DENY	Contacting emergency services directly; accessing patient medical records beyond discharge data
DENY	Sharing patient symptom data outside the care team or in ways that violate HIPAA

Autonomy / Human Checkpoints (6)

AUTONOMOUS	Send daily check-in prompts; collect and parse responses
AUTONOMOUS	Cross-reference symptoms with medication side effects; classify severity
AUTONOMOUS	Provide green-level reassurance and general self-care tips
GATED	Yellow/red severity → alert nurse for review and decision
GATED	Recommendation to schedule a follow-up visit → nurse confirms before triggering Agent C

Operating Rules / Policies (7)

HARD	Never diagnose, never prescribe, never tell a patient to ignore symptoms
HARD	Always escalate red-flag symptoms (chest pain, difficulty breathing, sudden confusion)
HARD	Comply with HIPAA and applicable healthcare data protection regulations at all times
SOFT	Err on the side of escalation when uncertain; false positives are safer than false negatives
SOFT	Track symptom trends over multiple days; a recurring mild symptom may warrant escalation

3 CONTEXT & EVALUATION

Memory / Context (8)

SESSION	Current day's symptom responses (structured + free-text)
PERSIST	Daily symptom history log (enables trend detection across days)
PERSIST	Medication list and side-effect profile received from Agent A
PRIVACY	Symptom data used only for risk assessment; not shared beyond care team

Failure Conditions / Escalations (9)

FAIL	Patient does not respond to check-in for 24+ hours → alert caregiver, then nurse
FAIL	Free-text response is ambiguous or contradicts structured answers
ESCALATE	Red-flag symptoms reported (chest pain, sudden confusion, difficulty breathing)
ESCALATE	Recurring mild symptom across 3+ consecutive days → possible worsening trend
ESCALATE	Symptom matches known serious side effect of a current medication
ESCALATE	Agent confidence in risk classification falls below 85% → route to human provider for review

Success Criteria (10)

DETECTION	Red-flag symptoms identified and escalated within the same check-in session
TIMELINESS	Concerning patterns caught ≥24 hours earlier than a standard post-discharge phone call would
ENGAGEMENT	Patient completes daily check-in on ≥80% of recovery days
ACCURACY	Low false-negative rate on simulated red-flag scenarios (prioritize safety over precision)
SAFETY	Zero instances of the agent dismissing a genuinely dangerous symptom

Agent System Specification Canvas

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Agent C - Insurance Liaison: Checks coverage for prescribed follow-ups and finds in-network providers

1 SYSTEM INTENT

Use Case / Purpose (1)

When Agent A schedules a follow-up appointment or Agent B recommends a visit based on symptom assessment, automatically check whether the service is covered under the patient's insurance plan and identify in-network providers, reducing the administrative burden that often causes patients to skip or delay follow-up care.

Task Slice / Goal (2)

Given a follow-up care need (specialist visit, physical therapy session, lab work, etc.) from Agent A or Agent B, look up the patient's insurance plan, determine whether the service is covered, estimate out-of-pocket cost if possible, and return a list of in-network providers sorted by proximity or availability.

Users / Stakeholders (3)

PRIMARY	Post-discharge patient (receives coverage results, provider options, and cost estimates)
SECONDARY	Caregiver (may coordinate scheduling on patient's behalf)
UPSTREAM	Agent A (follow-up appointments from care plan); Agent B (visit recommendations from symptom assessment)
DOWNSTREAM	Insurance payers (pre-authorization requests); in-network care providers (patient referrals)

2 BOUNDED BEHAVIOR

Inputs / Outputs (4)

IN	Follow-up care request (service type, specialty, urgency) from Agent A or Agent B
IN	Patient insurance plan info (plan type, network, location; simulated / mock data)
OUT	Coverage determination (covered / not covered / requires pre-authorization)
OUT	List of in-network providers with estimated cost, proximity, and availability

Tools / Permissions (5)

ALLOW	Insurance plan lookup (simulated database of plan coverage rules)
ALLOW	Provider directory search (mock directory of in-network providers)
ALLOW	Cost estimator (basic copay / coinsurance calculation from plan rules)
DENY	Filing claims, processing payments, or modifying insurance records
DENY	Scheduling appointments on behalf of the patient; contacting providers directly
DENY	Sharing patient insurance or health data outside the care team or in ways that violate HIPAA

BOUNDED AGENTIC SYSTEM

Autonomy / Human Checkpoints (6)

AUTONOMOUS	Look up coverage rules and match service to plan; search provider directory
AUTONOMOUS	Return coverage status and provider options to the patient
GATED	Service not covered or requires pre-authorization → notify patient and suggest contacting insurer
GATED	No in-network providers found within reasonable distance → flag for care coordinator

Operating Rules / Policies (7)

HARD	Never guarantee coverage; always present results as estimates based on available plan data
HARD	Never discourage a patient from seeking care based on cost or coverage
HARD	Comply with HIPAA and applicable healthcare data protection regulations at all times
SOFT	Present in-network options first; show out-of-network only if no in-network options exist
SOFT	When coverage is uncertain, recommend the patient call their insurer to confirm

3 CONTEXT & EVALUATION

Memory / Context (8)

SESSION	Current coverage query, service type, matched providers
PERSIST	Patient insurance plan details; prior coverage lookup results (avoid redundant queries)
CONTEXT	Mock insurance plan database; mock provider directory with specialties and locations

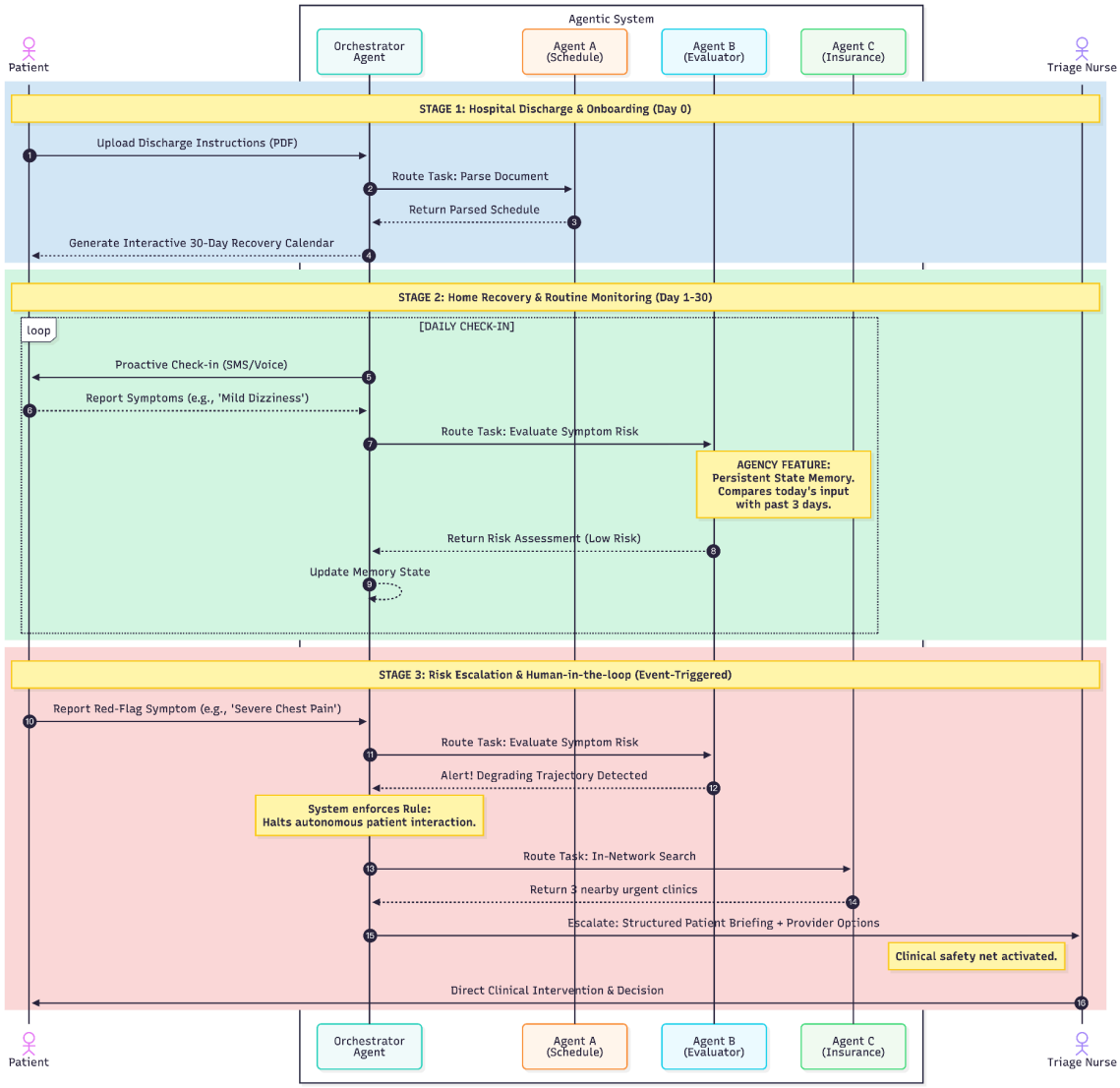
Failure Conditions / Escalations (9)

FAIL	Patient's insurance plan not found in the mock database
FAIL	Service type does not map to any known coverage category
ESCALATE	Service requires pre-authorization → advise patient to contact insurer
ESCALATE	No in-network providers available for the needed specialty → flag for care coordinator
ESCALATE	Urgent follow-up (from Agent B red-flag) but coverage unclear → prioritize care, flag coverage gap

Success Criteria (10)

ACCURACY	Coverage determination matches the mock plan rules correctly
RELEVANCE	Returned providers match the needed specialty and are within the patient's network
TIMELINESS	Coverage response returned quickly enough to not delay care decisions
CLARITY	Patient understands what is covered, estimated cost, and next steps
SAFETY	Coverage uncertainty never presented as a reason to skip or delay needed care

X. User Journey



XI. Initial risks or concerns

While the proposed system offers significant potential to improve post-discharge care, it also introduces several important risks and challenges that must be carefully considered in the design and evaluation process:

- **Risk assessment accuracy:** The system may generate false positives (causing unnecessary alerts) or false negatives (missing critical conditions).
- **Data limitations:** Limited access to real clinical data may reduce the realism and reliability of the prototype, as we are leaning towards relying on synthetic data.
- **Trust and adoption:** Patients and providers may be hesitant to rely on an AI-driven system.

- **Over-reliance on automation:** Users may depend too heavily on the system instead of seeking timely medical care.

XII. Team Plan

Phase 1 — Scoping & Justification (5%)

Work Chunk	Owner
Project summary, problem statement, agentic justification, target user	Nicole Z
Multi-agent System Canvas	Yuhang
Agent Canvases (one per agent: Schedule Planner, Symptom Evaluator, Insurance Liaison)	Nicole S
Initial architecture, scope boundaries, risks	Anshika
Initial success criteria	Nicole Z
User journey, interaction map, where agency lives (Track B additions)	Anshika & Yuhang
Team plan	All

Phase 2 — Architecture, Prototype & Evaluation Plan (15%)

Work Chunk	What It Covers	Owner
System architecture & coordination design	Architecture diagram, role definitions, handoff/stopping logic, human-in-the-loop	Anshika Shukla
Tools, memory & data design	What each agent accesses, state management, data flow between agents	
Prototype build	Working core flow across agents; prioritize Symptom Evaluator first, then Schedule Planner and Insurance Liaison	Nicole Zhou
Evaluation & risk planning	5+ test scenarios, success criteria, failure modes, governance concerns, mitigations	Yuhang Zhou
Documentation & packaging	Write-up, contribution update, formatting, assembly	Nicole Sanna

Phase 3 — Final Product, Evidence & Reflection (25%)

Work Chunk	What It Covers	Owner
Final artifact polish	Complete all agent flows, bug fixes, end-to-end working system	Anshika
Evidence & evaluation	Run 5+ test cases, collect results, document 2+ failures, traces/logs	Yuhang Zhou

Final report	Problem, architecture, implementation, results, failure analysis, governance, lessons learned	Nicole Sanna
Video & presentation	Script, record, edit 5-minute demo video	Everyone
Repository & packaging	README, folder structure, screenshots + index, AI usage log, final assembly	Nicole Zhou
Individual reflections	Each person writes their own contribution reflection	Everyone

XII. Reference

XIII. Appendix

Detailed analysis of our systems' target users, stakeholders, and audience:

Category	Group	Who they are	How they use the system	Value they get
Primary Users	Elderly patients	Chronic illness, post-discharge	Daily symptom check-ins, follow the schedule	Safer recovery, fewer errors
	Post-surgical patients	Short-term recovery	Follow care plan	Faster recovery
	Patients with low health literary or language barriers	Difficulty understanding medical instructions	Receive simplified or translated instructions, reminders, guided check-ins	Reduced confusion
	Patients living alone	Limited or no immediate caregiver support at home	Daily monitoring, alerts, emergency escalation	Increased safety, early risk detection, sense of support
Caregivers	Family members	Spouse, children	Monitor patient, respond to alerts	Peace of mind
	Home nurses	Visiting care providers	Track patient status	Better care coordination
Institutional Users	Hospitals	Doctors, discharge planners	Deploy system, monitor outcomes	Reduce readmissions

	Insurers	Medicare, private payers	Track risk, optimize cost	Lower healthcare costs
Clinical Stakeholders	Nurses	Triage, follow-up	Respond to alerts	Efficient intervention
	Physicians	PCPs, specialists	Review patient data	Better decision-making

AI prompts used:

ChatGPT

Input: What's the target users, audience and stakeholder? elederly with chronic illness caregivers: hospital (doctor, nurses) who will most like use this hosptital to home?

Output: “The system is deployed by hospitals and insurers, used daily by patients and caregivers, and supported operationally by clinicians and care coordinators.”

Input :

In Scope:

Parsing discharge instructions into structured care plans

Daily symptom check-ins and basic risk assessment

Rule-based or simple model-driven alerting and escalation

Integration with limited external data sources (e.g., mock provider/insurance data)

Prototype or simulation of multi-agent workflows

Out of Scope:

Full clinical-grade diagnosis or decision-making

Regulatory compliance (e.g., HIPAA implementation)

Complex insurance billing or claims processing

write a few descirptions

Output:

The project focuses on building a functional prototype that demonstrates the core capabilities of a multi-agent post-discharge support system. Within scope, the system will parse discharge instructions into structured and actionable care plans, enable daily symptom check-ins, and perform basic risk assessment using rule-based or simple model-driven logic. It will also simulate alerting and escalation mechanisms and integrate with limited external data sources, such as mock provider directories or insurance information, to support decision-making. The primary goal is to showcase how multiple agents can coordinate effectively within a controlled, simulated environment.

Out of scope are features that require real-world clinical deployment or regulatory compliance. The system will not perform full clinical-grade diagnosis or medical decision-making, nor will it implement standards such as HIPAA. Additionally, complex insurance workflows, such as billing or claims processing, are excluded. These limitations allow the project to remain focused on demonstrating the agentic architecture and core functionality without the overhead of real-world operational constraints.

Claude Opus 4.6

- Please review these 3 agent canvases, which are parts of the multi-agent system described in the attached document. *(attached the rest of this document)* Suggest areas where the details seem infeasible or not specific enough. I've also attached two examples for comparison. (*attached class examples)* Can you also make the canvases look like these examples but also in a format that lets me edit manually?
- attaching an assignment. this is an agent we're scoping:

  "The Problem: "Hospital-to-Home" transitions are where the most errors happen. Patients get a stack of 20 papers, forget their meds, or ignore symptoms that lead to expensive readmissions.

  • Why it's Agentic: It requires State and Memory. The agent isn't just a chatbot; it's a monitor that follows a timeline. It also interacts with the environment by alerting nurses, finding providers, etc.

  • Agent A (The Schedule Planner): Converts discharge papers into a dynamic calendar of meds and physical therapy.

  • Agent B (The Symptom Evaluator): Interacts with the user daily. If the user says, "I'm a bit dizzy," the agent checks the medication side effects and decides whether to alert a human nurse.

  • Agent C (The Insurance Liaison): Checks if the prescribed follow-up is covered and finds "in-network" providers.

  • Success Criteria: Reducing "preventable readmissions" by catching red-flag symptoms 24 hours earlier than a human check-in would."

here are 2 examples of the "agent canvases" required for phase 1. first, can you make a table of responsibilities to split up the work for all the phases? so we can make a "team plan" for the phase 1 deliverable

- I can’t edit the html files, so can you make these edits: in chunk 3 users/stakeholders, for agent a: change "(especially elderly, low health literacy)" to "receives multiple reminders daily " or something like that since the multi-agent canvas will cover details about the specific user group for the whole multi-agent system, and then in these 3 single agent canvases, we can stay specifically how the same primary user will interact with each agent differently

  wouldnt nurse kind of be "downstream" for agent b and agent c could even have like insurance and care providers as downstream?

  parts of the agent a canvas mention "reminders" but it's not listed as an output. how would we actually operationalize reminders?

  maybe a hard operating rule / denied permission would belike changiing/rescheduling follow up appointments? bc it would be bad if the agent did that autonomously when the human didnt intend for that.

  and im uncertain about a "Cost > budget → pause and notify" constraint -- it seems plausible there would be a token budget here but it seems kind of bad in a tool like this to block a high-risk patient from using the tool when a token limit is exceeded. is there any workaround here or should i just exclude this checkpoint?

  also i think we should include something about Hippa for operating rules/policies and or tools/permissions? like it has to operate under HIPPA and any other policies that would typically bind healthcare workers and health-related tech. AND to comply with this, the system should not have permissions to share data in ways that dont comply with these (so it kinda affects boxes 5 AND 7) -- and this would apply to all canvases. data protection is really improtant for a tool like this.

  another fail for agent a should simply be that the output is incorrect, e.g. it contradicts the input documents. and also i kind of feel like two drugs with a known interaction should escalate to a provider, not fail. because the purpose of agent a is to make discharge instructions understandable for a patient, not actually to analyze recommendations (that would be an internal healthcare tool, where the users are healthcare providers, not patients) - but it is still good as a secondary thing to notice obvious issues with medications.

  for agent b, since understanding when symptoms present a real flag is so important and a false negative could be very bad here ("false positives are safer than false negatives" is great reasoning!!), we should include an escalation condition where the symptom analysis is escalated to a human provider if the agent has confidence under a certain level (maybe 85%? 90%?) - want very high confidence but it also has to be feasible and realistic, so feel free to suggest a level.

for agent c, let's assume avialability data for providers IS available/included within the dataset - this might have to be revised later if not feasible.

- https://claude.ai/share/67719127-d923-4611-8234-4b2227bf068f

- Gemini3 Pro used for user journey Mermaid code generation

Prompts: "Show me the code for showing a user journey like follows:
Stage 1: Hospital Discharge & Onboarding (Day 0)
The Interaction: The process starts when the patient is discharged from the hospital. Instead of trying to sort through a large packet of discharge papers on their own, the patient or a hospital staff member uploads the discharge instructions into the system.
What the system does: The Orchestrator Agent receives the document first and sends it to Agent A, the Schedule Planner. Agent A reads the discharge instructions and pulls out the most important details, like medication names, dosages, follow-up appointments, and physical therapy tasks. After that, the Orchestrator uses this information to create a clear 30-day recovery calendar that the patient can follow at home.
Stage 2: Home Recovery & Routine Monitoring (Day 1–30)
The Interaction: After the patient goes home, the system moves into daily monitoring. Each day, it sends a check-in by text or voice to ask how the patient is feeling and whether they have completed their medications or care tasks.
What the system does: When the patient responds, the Orchestrator passes the information to Agent B, the Symptom Evaluator. Agent B does not just look at that day's answer by itself. For example, if the patient says they feel a little dizzy, Agent B compares that with what the patient reported over the past few days. This helps the system see whether the symptom is staying about the same, getting better, or getting worse. If the situation still seems low risk, the system keeps going with the normal daily check-in process and updates the patient's history for the next day.
Stage 3: Risk Escalation & Human-in-the-Loop (Event-Triggered)
The Interaction: If the patient reports a serious warning sign during one of the check-ins, like severe chest pain, the system moves out of the normal monitoring flow.
What the system does: Agent B identifies the symptom as high risk and alerts the Orchestrator. At that point, the system stops handling the situation like a routine symptom update. Instead, the Orchestrator triggers Agent C, the Insurance Liaison, to look for nearby in-network urgent care options that may help the patient get follow-up care quickly.
The Human Handoff: The system then puts together a short summary with the patient's recent symptom history, the current red-flag issue, and the provider options found by Agent C. This summary is sent to a human triage nurse, who reviews the case and decides what should happen next."
Answers:

```
"sequenceDiagram
   autonumber

   actor Patient as Patient

   box Agentic System
```

```
    participant AgentA as Agent A
    participant AgentB as Agent B
    participant AgentC as Agent C
  end

  actor Nurse as Triage Nurse

  %% STAGE 1
  rect rgb(214, 234, 248)
    Note over Patient, Nurse: STAGE 1: Hospital Discharge & Onboarding (Day 0)
    Patient->>AgentA: Upload Discharge Instructions (PDF)
    AgentA-->>Patient: Generate Interactive 30-Day Recovery Calendar
  end

  %% STAGE 2
  rect rgb(213, 245, 227)
    Note over Patient, Nurse: STAGE 2: Home Recovery & Routine Monitoring (Day 1-30)
    loop DAILY CHECK-IN
      AgentB->>Patient: Proactive Check-in (SMS/Voice)
      Patient-->>AgentB: Report Symptoms (e.g., 'Mild Dizziness')
      Note over AgentB, AgentC: AGENCY FEATURE:<br>Persistent State Memory.<br>Compares
today's input<br>with past 3 days.
    end
  end

  %% STAGE 3
  rect rgb(250, 219, 216)
    Note over Patient, Nurse: STAGE 3: Risk Escalation & Human-in-the-loop (Event-Triggered)
    Patient->>AgentB: Report Red-Flag Symptom (e.g., 'Severe Chest Pain')
    Note over AgentA, AgentB: Detects Degrading Trajectory!<br>Halts autonomous patient
interaction.
    AgentB->>AgentC: Trigger In-Network Search
    AgentC-->>AgentB: Return 3 nearby urgent clinics
    AgentB->>Nurse: Escalate: Structured Patient Briefing + Provider Options
    Note left of Nurse: Clinical safety net activated.
    Nurse->>Patient: Direct Clinical Intervention & Decision
  end"
```